

BASE MARAMBIO, Antarctica

WMO: 890550

Lat: 64.241S

Lon: 56.625W

Elev: 200

StdP: 98.95

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-29.5	-27.6	-32.7	0.2	-28.8	-30.6	0.2	-27.2	30.6	-18.6	28.0	-22.1	13.5	200	1.259

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.6	7.5	3.8	6.1	2.9	4.8	2.0	4.2	6.8	3.2	5.5	2.3	4.3	4.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2.1	4.5	4.3	1.2	4.2	3.2	0.5	4.0	2.4	17.6	7.0	15.8	5.6	14.2	4.4	9.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.2	22.4	20.0	DB	-30.9	10.5	2.6	1.9	-32.7	11.9	-34.3	13.0	-35.7	14.0	-37.6	15.4	(4)
(5)			WB	-30.9	6.7	2.1	1.6	-32.5	7.8	-33.7	8.7	-34.9	9.6	-36.5	10.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.0	-0.8	-1.7	-5.2	-10.5	-11.5	-14.8	-15.1	-13.4	-10.7	-7.1	-3.6	-1.4	(6)
(7)		DBStd	8.10	2.38	3.55	5.46	7.15	7.45	7.93	8.19	7.75	8.01	6.41	3.99	2.52	(7)
(8)		HDD10.0	6573	335	327	471	614	666	744	777	725	622	531	409	352	(8)
(9)		HDD18.3	9614	593	560	729	864	924	994	1036	983	872	789	659	611	(9)
(10)		CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	8.4	6.3	7.1	8.1	8.7	8.4	9.5	9.3	9.2	9.5	8.9	8.2	7.3	(14)
(15)	Precipitation	PrecAvg	851	63	70	92	75	68	53	60	67	77	90	78	59	(15)
(16)		PrecMax	1106	92	102	112	132	117	95	107	130	119	135	121	99	(16)
(17)		PrecMin	588	37	46	51	20	31	21	14	11	37	59	37	22	(17)
(18)		PrecStd	152	16	15	15	25	21	17	25	29	26	20	23	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	9.2	7.5	6.1	8.0	3.9	4.5	3.4	5.3	6.1	8.6	7.4	(19)
(20)			MCWB	5.6	5.1	4.0	2.2	3.6	0.0	0.5	0.2	1.4	2.4	4.7	3.7	(20)
(21)		2%	DB	6.5	6.9	5.5	3.2	4.3	1.5	1.2	1.1	3.1	3.9	5.4	5.2	(21)
(22)			MCWB	3.6	3.5	2.5	0.7	1.2	-0.4	-0.8	-1.0	0.5	1.2	2.5	2.4	(22)
(23)		5%	DB	4.7	5.2	3.8	1.7	2.1	-0.2	-0.7	-0.6	1.7	2.6	3.3	3.8	(23)
(24)			MCWB	2.4	2.4	1.5	0.1	-0.1	-1.4	-1.8	-1.7	0.0	0.5	1.2	1.5	(24)
(25)		10%	DB	3.0	3.7	2.2	0.1	-0.1	-1.9	-2.1	-2.0	0.3	1.4	1.8	2.5	(25)
(26)			MCWB	1.4	1.5	0.7	-0.9	-1.0	-2.7	-2.9	-2.8	-0.8	-0.1	0.3	0.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	5.4	4.6	2.7	3.8	0.7	1.0	0.9	2.2	3.0	4.9	4.2	(27)
(28)			MCDB	8.2	8.7	6.8	5.4	7.6	2.1	3.2	2.5	4.2	5.4	8.6	6.7	(28)
(29)		2%	WB	3.8	3.9	2.9	1.2	1.6	-0.3	-0.5	-0.6	0.9	1.6	2.8	2.7	(29)
(30)			MCDB	6.1	6.4	5.1	2.7	3.8	1.3	0.9	0.5	2.5	3.3	5.1	4.7	(30)
(31)		5%	WB	2.6	2.7	1.7	0.1	0.1	-1.3	-1.7	-1.7	0.1	0.7	1.4	1.8	(31)
(32)			MCDB	4.3	4.9	3.4	1.5	1.6	-0.3	-0.7	-0.7	1.4	2.2	3.0	3.5	(32)
(33)		10%	WB	1.5	1.7	0.7	-1.0	-1.1	-2.7	-2.9	-2.8	-0.8	-0.1	0.3	0.9	(33)
(34)			MCDB	2.9	3.5	2.1	0.2	0.0	-1.8	-2.0	-1.9	0.4	1.3	1.6	2.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.6	4.3	5.3	6.2	7.1	6.4	7.5	7.3	7.3	5.9	4.6	3.7	(35)
(36)			MCDBR	6.5	7.0	6.9	7.9	7.4	7.7	8.5	8.4	7.6	6.5	6.6	6.0	(36)
(37)			MCWBR	4.5	4.9	5.1	6.3	5.3	6.7	7.5	7.3	6.0	4.9	4.9	4.1	(37)
(38)		5% WB	MCDBR	6.3	6.7	6.5	8.0	7.5	7.7	8.5	8.4	7.1	6.3	6.2	5.7	(38)
(39)			MCWBR	4.6	5.0	5.1	7.0	6.2	7.0	7.8	7.7	6.1	5.1	4.8	4.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.278	0.272	0.258	0.225	0.152	0.139	0.142	0.213	0.244	0.262	0.266	0.275	(40)	
(41)		taud	2.440	2.453	2.454	2.331	1.993	1.933	1.956	2.243	2.306	2.326	2.375	2.373	(41)	
(42)		Ebn at Noon	994	948	872	715	467	220	461	709	864	947	1000	1008	(42)	
(43)		Edh at Noon	101	88	69	46	22	9	21	48	77	98	106	111	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.26	4.47	2.52	1.03	0.25	0.06	0.15	0.73	2.23	4.35	6.37	7.23	(44)	
(45)		RadStd	0.64	0.42	0.31	0.09	0.02	0.00	0.01	0.06	0.12	0.17	0.32	0.71	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page